

A CNN-TRANSFORMER FUSION MODEL-BASED EEG SIGNAL-DRIVEN METHOD FOR GAME VIDEO EMOTION CLASSIFICATION

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Received 13 January 2025

Accepted 7 May 2025

Published 7 August 2025

Affective computing is a vital research area in artificial intelligence and human–computer interaction, with wide-ranging applications in brain–computer interfaces (BCIs) and multi-modal emotion analysis. To achieve emotion classification in game videos, this paper proposes a deep feature fusion-based model that accurately recognizes emotional states from electroencephalography (EEG) data. The model adopts a dual-branch architecture combining convolutional neural networks (CNNs) and Transformers to extract both local and global features from EEG signals. The CNN branch utilizes depth-wise separable convolution modules to efficiently capture time-frequency features while reducing computational cost. In parallel, the Transformer branch employs a multi-head self-attention (MHSA) mechanism to model global dependencies, uncovering long-range emotional correlations. Features from both branches are fused and passed through global pooling and fully connected layers for classification. Experimental results show that the proposed model excels in emotion recognition tasks for game videos, enhancing EEG feature modeling and improving classification accuracy and generalization. By jointly capturing local details and global patterns, the model effectively identifies complex emotional dynamics. In the context of game interaction, it can perceive players' emotional changes and support real-time game content adaptation. This research offers an efficient deep learning framework for EEG-based emotion analysis, with promising applications in personalized entertainment, intelligent game design, and mental health monitoring. It establishes a foundation for real-time affective interaction and paves the way for future research on cross-modal emotional fusion and adaptive feedback mechanisms in immersive environments.

Keywords: Sentiment computing; EEG; CNN; transformer; emotion classification.

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1. Introduction

In fast-paced and immersive gaming environments, players' emotional states often fluctuate rapidly in response to changing stimuli, which creates challenges for real-time emotion-aware adaptation. Accurately identifying these subtle and dynamic emotional shifts is crucial for enabling intelligent game systems that can personalize content, adjust difficulty, and enhance engagement. Emotion analysis — especially based on physiological signals such as electroencephalography (EEG) — offers a promising solution by providing deeper insights into players' internal affective states, beyond what is observable through facial expressions or voice.¹ With advances in artificial intelligence and brain-computer interface (BCI) technology, EEG-based emotion recognition has become an emerging area of interest in affective computing. EEG signals, due to their high temporal resolution and strong emotional correlation, are increasingly used to monitor users' mental states in real time. Compared to traditional modalities, EEG is less susceptible to surface-level distractions and environmental noise, making it more suitable for complex interactive scenarios such as gaming. In the context of game design, EEG-based emotion analysis enables systems to sense player states and dynamically adjust game elements to maintain optimal engagement. For instance, when boredom is detected, games may introduce new stimuli; when anxiety rises, the system can moderate challenges to reduce stress. These emotion-driven adaptations not only enhance immersion but also pave the way for personalized and emotionally resonant gaming experiences. Therefore, efficient modeling of EEG signals for emotion classification holds both theoretical significance and practical value for the development of next-generation intelligent and adaptive gaming systems.

In recent years, researchers have widely employed convolutional neural networks (CNNs) to extract spatial features from EEG data for the purpose of emotion classification. Nunez *et al.* introduced a CNN-LSTM model that integrates CNN for local spatial feature extraction with LSTM for temporal dependency capture, facilitating the extraction of spatiotemporal information from EEG data.² Although this method achieves joint modeling of local features and temporal dependencies, the modeling of these features remains relatively independent, lacking deep integration and limiting its ability to capture global features. Zhao *et al.* presented a multi-level feature-guided capsule network utilizing multi-channel EEG, wherein CNNs extract local time-frequency features, and the Capsule Network models the spatial relationships of EEG signals.³ However, this method primarily focuses on local and spatial features and fails to effectively capture the global dependencies of EEG signals, restricting its capacity to model complex emotional patterns. Kim *et al.* introduced an enhanced LSTM model to derive temporal features from EEG data for the purpose of emotion recognition.⁴ While this model excels in temporal dimension modeling, its depth of EEG spatial feature extraction is limited, and it does not incorporate global contextual information, presenting certain limitations. As the advantages of Transformer models in global feature modeling (GFM) have become widely recognized,

they have gradually been applied in EEG-based emotion recognition. Guo *et al.* introduced a model that integrates deep convolution with a Transformer encoder for the categorization of emotions using EEG data.⁵ This method leverages the self-attention mechanism of Transformer to capture global dependencies but performs inadequately in handling fine-grained local features. Due to the detailed information contained in EEG time-frequency representations, relying solely on Transformer may result in missing critical features. In summary, existing studies present certain limitations in modeling both local and global features, particularly in the deep integration of local and global features, capturing complex emotional patterns, and effectively utilizing the multi-dimensional characteristics of EEG signals, which require further optimization.

This work proposes a dual-branch emotion classification model that combines CNN and Transformer to enhance local and GFM for emotion recognition using EEG signals, addressing shortcomings identified in prior research. The model utilizes the synergistic function of CNN and Transformer to extract and integrate local time-frequency characteristics and global dependencies from EEG data, hence improving the precision of emotion classification. The precise contributions are enumerated as follows:

- (1) This paper introduces a dual-branch architecture that integrates a local feature extraction (LFE) Module and a GFM module. LFE is designed to extract local time-frequency features from EEG time-frequency representations, while GFM captures global contextual dependencies through the self-attention mechanism. This design effectively overcomes the singularity in local and GFM found in existing models, enhancing the ability to describe complex emotional patterns.
- (2) A dedicated Fusion Block is introduced to achieve bidirectional information connection and aggregation between LFE and GFM. This strategy dynamically integrates local and global features, strengthening the model's feature representation capacity and further improving classification performance.
- (3) The proposed model was extensively evaluated on the DEAP dataset. Results show that it improves classification accuracy in the valence dimension (Valence-Acc) by 7.16% and in the arousal dimension (Arousal-Acc) by 5.82% compared to the baseline CNN model. These performance gains highlight the benefit of integrating local and global feature representations, as well as the effectiveness of the proposed feature fusion strategy. The findings support the potential of this approach for advancing EEG-based emotion classification and contribute new perspectives to the modeling of affective signals.

2. Related Knowledge

2.1. EEG signals and their relationship to sentiment analysis

EG signals, as electrophysiological reflections of brain activity, exhibit complex and dynamic characteristics that pose challenges for effective analysis.⁶ One key property

is their nonstationarity — EEG signals are time series whose statistical properties change continuously over time, reflecting pronounced temporal variability. This makes it difficult for traditional linear models to capture such signals, necessitating time-frequency analysis (such as short-time Fourier transform (STFT)⁷ and continuous wavelet transform⁸) to extract dynamic features from the signals. Second, EEG signals possess high temporal resolution, typically recorded at sampling rates ranging from 128 Hz to 1024 Hz, which allows for real-time reflection of neural dynamics and provides critical data support for affective computing tasks. However, this high temporal resolution also poses challenges in processing the large volumes of data generated. Third, the low signal-to-noise ratio (SNR) of EEG signals is another key characteristic, as they are highly susceptible to artifacts and environmental noise, affecting signal reliability. Thus, techniques such as filtering and independent component analysis (ICA)⁹ are required to clean the signals. Finally, the multi-channel nature of EEG signals adds to their complexity. Each channel records activity from different brain regions, reflecting interactions between regions and overall brain dynamics. However, the multi-channel data increases modeling difficulty, often necessitating deep learning models to extract and integrate spatial features across channels.¹⁰ In summary, these characteristics make EEG signals an essential data source for emotion analysis, while simultaneously posing technical challenges for efficient modeling. Table 1 presents the relationship between EEG frequency bands and emotional states.

Figure 1 illustrates the distribution of EEG frequency bands within the valence–arousal space. This illustration is based on a two-dimensional (2D) emotion model and visually depicts the relationship between EEG frequency components and emotional states. The X-axis denotes emotional valence, ranging from low valence (e.g., negative emotions) to high valence (e.g., positive emotions), while the Y-axis represents emotional arousal, from low (e.g., calm moods) to high (e.g., excitement).

As shown in Fig. 1, δ and θ waves are located in the lower-left region, representing low valence and low arousal states, typically associated with deep relaxation or meditation. α waves appear slightly to the left of the center, indicating moderate

Table 1. Correspondence between EEG frequency bands and emotional states.

EEG band	Frequency range (Hz)	Correlation with emotional states	Associated brain regions
Delta (δ)	0.5–4 Hz	Related to deep sleep and low-arousal emotions	Brainstem, Thalamus
Theta (θ)	4–8 Hz	Associated with relaxation, meditation, and mild focus	Frontal and Temporal lobes
Alpha (α)	8–12 Hz	Linked to relaxation and positive emotions (low arousal)	Occipital, parietal, and temporal lobes
Beta (β)	13–30 Hz	Related to anxiety, heightened alertness, and tension	Prefrontal cortex, motor cortex
Gamma (γ)	> 30 Hz	Associated with focus, high arousal, and intense emotional states	Frontal, temporal, and parietal lobes

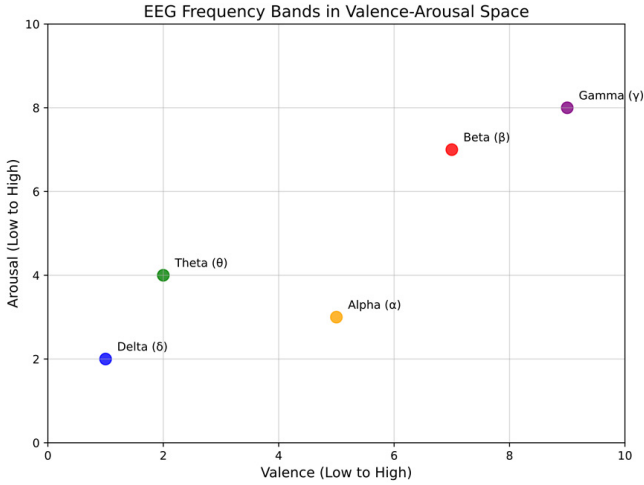


Fig. 1. EEG frequency bands in valence–arousal space.

valence and low arousal, often linked to relaxed and pleasant emotions. β and γ waves are positioned in the upper-right region, reflecting high valence and high arousal, commonly associated with intense emotional responses such as tension, focus, or excitement. This spatial distribution reveals notable variations in emotional valence and arousal levels across EEG frequency bands and offers useful insights into how these components correlate with affective states.

2.2. Existing EEG-based emotion classification methods

While various approaches have been proposed for EEG-based emotion classification, most existing methods struggle to simultaneously and effectively model both local time-frequency features and global contextual dependencies. To address this gap, our work proposes a dual-branch classification model that integrates CNNs and Transformer architectures, aiming to leverage CNN's strength in extracting local features and Transformer's ability to model global relationships. In recent years, researchers have thoroughly investigated the integration of CNN and its enhanced models for the extraction of spatiotemporal characteristics from EEG signals. For instance, Chowdary *et al.* utilized a single RNN structure focused on temporal dependency modeling of EEG signals, simplifying model complexity but with insufficient spatial feature extraction.¹¹ Similarly, Meng *et al.* proposed a cascaded convolutional recurrent neural network (CRNN), which extracts spatial features using CNN and captures temporal dependencies with RNN, achieving precise recognition of emotional states from EEG signals.¹² Moreover, Zhong *et al.* integrated TQWT feature extraction with a hybrid CRNN model, enhancing classification performance by preprocessing to improve signal characteristics and extracting multi-dimensional features.¹³ These approaches leverage the strengths of CNN and RNN simultaneously but may face high computational complexity when processing high-dimensional EEG data. Transformer

models, known for their powerful GFM capabilities, have gradually become a research hotspot in EEG-based emotion analysis. Wang *et al.* introduced a Transformer model that utilizes hierarchical spatial information learning to capture long-term relationships in EEG signals.¹⁴ Sharma *et al.* further combined Transformers with spatio-temporal modeling, demonstrating their applicability in visual brain decoding tasks.¹⁵ Cheng *et al.* designed a hybrid model with multi-scale dynamic CNNs and gated Transformers, enabling dynamic modeling at different feature levels.¹⁶ These studies highlight that Transformers significantly enhance global dependency modeling but may fall short in capturing local features, particularly for fine-grained information in complex emotional patterns.

To overcome the limitations of single models, researchers have explored hybrid models. Liu *et al.* developed a combined model integrating spatiotemporal Transformers with CNN-LSTM, retaining the global feature capabilities of Transformers while incorporating the local modeling strengths of CNNs and LSTMs.¹⁷ Chakravarthi *et al.* proposed a hybrid CNN-LSTM model that combines CNN's spatial feature extraction capabilities with LSTM's temporal dependency modeling, achieving accurate classification of emotional states from EEG signals. This approach performed well on multi-channel EEG data but introduced additional computational costs due to its complex hybrid architecture.¹⁸ Narsimha Reddy *et al.* adopted a hybrid feature fusion strategy and adaptive Transformer method, achieving more efficient emotion classification by modeling both temporal and frequency characteristics of EEG signals.¹⁹ Additionally, Kulkarni and Dixit proposed an improved hybrid classifier that enhances model performance by optimizing the feature set.²⁰ These hybrid methods improve emotion classification by integrating the strengths of multiple models but often come with increased design complexity and training costs, especially when dealing with high-dimensional EEG data.

In summary, EEG-based emotion classification methods have shown diverse development trends in recent years. CNN and its improved models excel in local spatial feature extraction but have limited capabilities for temporal dependency modeling. RNN and LSTM methods are proficient in time series modeling but lack depth in capturing spatial features. Transformer models are gaining attention for their robust GFM abilities, though their fine-grained modeling of local features remains insufficient. Hybrid models significantly enhance emotion classification performance by combining the advantages of different models but are often computationally expensive and complex to design. This work introduces a dual-branch model that integrates CNN and Transformer to efficiently collaborate in modeling local and global data, thereby enhancing the accuracy and generalization of EEG-based emotion categorization.

3. Proposed Model

3.1. Model architecture

This research presents a dual-branch emotion classification model that integrates CNN and Transformer for EEG-based emotion classification. The model is

engineered to optimize the LFE capabilities of CNN and the global dependency modeling strengths of Transformer, facilitating the integrated modeling of both local and global characteristics. The fundamental premise of the model is to extract multi-level characteristics from EEG data using a dual-branch architecture. A feature fusion module dynamically amalgamates information from both branches to improve the precision and generalization of emotion classification. Figure 2 depicts the architecture of the suggested model.

The suggested model comprises the following modules:

Input Module. The input comprises the time-frequency representation of EEG signals. This work used the STFT to transform raw time-series signals into a 2D picture format for further feature extraction. Compared to wavelet transforms, STFT offers simpler implementation and uniform time-frequency resolution, which is advantageous in real-time applications. Given the relatively stable sampling rate and medium-scale temporal resolution required in our task, STFT was chosen as a more practical and efficient solution. The input dimensions are $C \times H \times W$ (channels $\times$ image height $\times$ image width).

LFE Module. This module consists of three CNN blocks. Each block includes depth-wise separable convolutions, batch normalization (BatchNorm), an activation function (ReLU), and optional residual connections. It extracts local time-frequency features from EEG spectrograms, capturing fine-grained patterns. The output is the locally extracted features, layer by layer, which are later fused with the global features from the Transformer branch.

GFM Module: This module comprises three Transformer blocks. Each block includes a multi-head self-attention (MHSA) mechanism, a feed-forward neural network (FFN), layer normalization (LayerNorm), and residual connections. Using self-attention, this module captures global dependencies in EEG signals, modeling long-range and cross-channel relationships. The output is the globally modeled features.

Fusion Block: This block integrates the output features from the CNN and Transformer branches, achieving dynamic collaborative modeling of local and global features. Through a bidirectional information flow mechanism, it merges the features from both branches, retaining local details and global context information.

Classification Module: The integrated features undergo dimensionality reduction using a global pooling layer and are subsequently processed through a fully connected layer to finalize the classification task. The result is the categorizing of emotions.

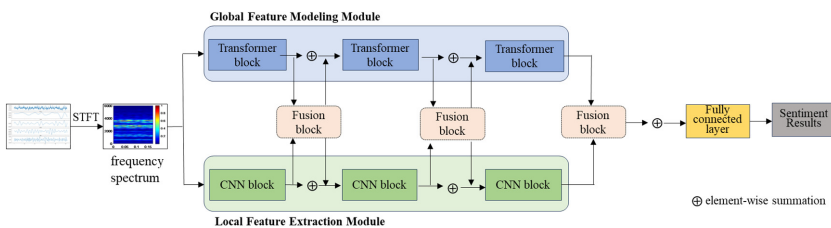


Fig. 2. Proposed model structure.

3.2. LFE module

The LFE Module is engineered to effectively extract hierarchical temporal-frequency characteristics from EEG spectrograms utilizing three CNN blocks. Each block leverages depth-wise separable convolution to reduce computational complexity while retaining critical local feature information. The detailed structure and functionality of each block are shown in Table 2.

The LFE module consists of three hierarchical CNN blocks, each designed to progressively extract temporal-frequency features from EEG spectrograms. The first block captures low-level patterns, while the second block focuses on mid-level features, and the third block extracts high-level local features crucial for classification. depth-wise separable convolutions are utilized in all blocks to reduce computational complexity while maintaining feature extraction efficiency. Residual connections in the first two blocks enhance gradient flow and stabilize training, ensuring effective learning of local features at each level. These extracted features provide a rich representation of local temporal-frequency details, serving as a critical input for the subsequent feature fusion with the global modeling branch.

3.3. GFM module

The GFM Module is designed to capture global dependencies and long-range relationships from EEG data using three Transformer blocks. Each block employs MHSA to extract dependencies across different temporal and spatial dimensions, complemented by FFNs to refine features. LayerNorm ensures stable training, while residual connections prevent gradient vanishing and improve convergence. The detailed structure and functionality of each block are shown in Table 3.

The GFM module uses a hierarchical structure with three Transformer blocks, each contributing progressively to the extraction of global features. The first block focuses on low-level dependencies, capturing fundamental relationships across temporal and spatial dimensions in the EEG data. The second block builds on this by extracting mid-level global features, identifying more complex interactions. The third block, targeting high-level dependencies, provides abstract and comprehensive global representations essential for the downstream classification task.

Table 2. LFE module structure.

Layer	Components	Function
CNN Block 1	Depth-wise Separable Convolution + BatchNorm + ReLU + Residual Connection	Extracts low-level temporal-frequency features
CNN Block 2	Depth-wise Separable Convolution + BatchNorm + ReLU + Residual Connection	Extracts mid-level temporal-frequency feature
CNN Block 3	Depth-wise Separable Convolution + BatchNorm + ReLU	Extracts high-level local features for classification

Table 3. Global feature extraction module structure.

Layer	Components	Function
Transformer Block 1	MHSA + FFN + LayerNorm + Residual Connection	Extracts low-level global dependencies from EEG data
Transformer Block 2	MHSA + FFN + LayerNorm + Residual Connection	Extracts mid-level global features
Transformer Block 3	MHSA + FFN + LayerNorm + Residual Connection	Extracts high-level global dependencies

The use of MHSA enables the model to effectively identify and represent long-range dependencies in EEG signals, crucial for understanding the intricate patterns associated with emotional states. FFN further enhances the feature extraction process by introducing nonlinear transformations. LayerNorm stabilizes the training process by normalizing activations across layers, and residual connections ensure robust gradient flow, improving model stability and performance.

The features extracted by the GFM module serve as a critical input for the Fusion Block, where they are integrated with the local features extracted by the LFE Module. This collaboration between local and global branches enhances the model's ability to capture both fine-grained and holistic patterns in EEG data, leading to more accurate and robust emotion classification.

3.4. Fusion module

The fusion module is one of the core components of the model, designed to dynamically integrate the output features from the LFE module and the GFM module. This module effectively combines local details with global context, enhancing feature representation and providing richer support for the final emotion classification. Feature fusion is performed using element-wise addition, merging the output features from the two branches as follows:

$$\alpha \cdot \beta \cdot F_{\text{fusion}} = F_{\text{global}} + F_{\text{local}}, \quad (1)$$

where F_{local} and F_{global} are the output features of the LFE Module and GFM Module, respectively, while represents the fused features. α and β are fusion weights learned through the attention mechanism.

The fused features are transformed using a nonlinear activation function (e.g., ReLU) to enhance the model's nonlinear representation capability. After the feature fusion, LayerNorm is introduced to ensure feature distribution stability and improve training efficiency.

4. Experiments and Analysis

4.1. Dataset and preprocessing

In the field of affective computing, particularly for emotion classification tasks involving game videos, EEG signals serve as crucial biosignals that reflect emotional

states, demonstrating unique advantages. To accurately identify emotional states, selecting high-quality public datasets is essential. The DEAP dataset, a benchmark multimodal affective analysis dataset, offers abundant EEG data, peripheral physiological signals, and emotion labels, laying a solid foundation for multimodal feature extraction and classification tasks. Therefore, this study adopts the DEAP dataset as the primary data source.

The DEAP dataset comprises 32 healthy volunteers (ages 19–37, gender-balanced), each of whom viewed 40 music videos exhibiting emotional attributes. Their physiological responses were documented with a 32-channel EEG and 8-channel peripheral physiological signals at a sampling frequency of 512 Hz. Each trial had 3 s of baseline signals followed by 60 s of stimulated signals. Following the viewing of the movies, participants evaluated them according to four dimensions: arousal, valence, dominance, and preference. This rigorous experimental design establishes the DEAP dataset as a vital resource in emotion analysis research.

To adapt the dataset for the proposed model, the preprocessing procedure involves the following steps: (1) Downsampling the EEG data to 128 Hz to significantly reduce computational complexity. (2) Artifact removal using ICA to eliminate ocular and other noise artifacts, followed by baseline correction to remove drift. (3) Applying a band-pass filter (1–47 Hz) to retain frequency components relevant to emotions, filtering out high-frequency noise and DC components.

The preprocessed EEG data preserves the entire 60 s signal from each trial, avoiding segmentation that could introduce label uncertainty and ensuring the accuracy of emotional labels. Each participant produces 40 signal samples, totaling 1280 segments, providing a solid foundation for further analysis. To satisfy the input criteria of multimodal models, the preprocessed EEG data is further converted into 2D time-frequency representations. The STFT converts time-series signals into spectrograms, adjusted to a range of [0,1], so offering standardized input for the CNN component. The STFT was applied with a 512-point Hamming window and 50% overlap at a sampling rate of 128 Hz, corresponding to a 4 s window. Although relatively long, this configuration offers high frequency resolution and stable spectrograms, which are beneficial for modeling sustained emotional states. STFT is effective in capturing time-varying frequency patterns relevant to emotion shifts in game video contexts.

In emotion classification, the target labels are categorized into four groups according to arousal and valence ratings. A threshold of 5 is employed to categorize emotions into four distinct classes: High arousal — high valence; High arousal — low valence; Low arousal — low valence; Low arousal — high valence. This classification delineates explicit training objectives for the model. A threshold of 5 is employed to categorize emotions into four distinct classes. . . This threshold follows common practice in DEAP-based emotion classification studies, as the self-assessment scores range from 1 to 9, with 5 being the neutral midpoint. It effectively separates high and low emotional intensities.

The dataset is randomly split into training, validation, and test sets at a ratio of 8:1:1, resulting in 1,024 (80%), 128 (10%), and 128 (10%) samples, respectively.

These are used for model training, parameter tuning, and performance evaluation. To enhance model generalization, data augmentation techniques are applied to the spectrograms, including temporal jittering, frequency perturbation, and translation operations, enriching the diversity of the samples. This systematic preprocessing pipeline ensures data quality and balance, providing high-quality, structured data to support subsequent research.

4.2. Experimental setup

A number of experiments were conducted utilizing the DEAP dataset to assess the performance of the proposed model. The model’s hyperparameters and structural configurations were carefully tuned to ensure optimal performance. The parameter settings for each module are summarized in Table 4.

In this study, the model’s overall performance is measured by Accuracy. This metric shows what percentage of EEG samples were correctly classified relative to all samples. It is determined by dividing the total number of samples by the number of samples that were correctly classified.

The hardware environment for the experiments includes an Intel Xeon E5-2698 v4 @ 2.20 GHz CPU, an NVIDIA Tesla V100 GPU (32 GB VRAM), 128 GB DDR4 RAM, and 1 TB NVMe SSD storage. The software environment is based on the Ubuntu 20.04 LTS operating system, utilizing the PyTorch 2.0 deep learning framework and Python 3.9 for model development. Auxiliary tools such as Scikit-learn are used for computing evaluation metrics, and Matplotlib is employed for result visualization. To ensure reproducibility of the experimental results, a fixed random seed (set to 42) was used during the experiments. This guarantees consistency in dataset splitting, weight initialization, and randomness during the training process.

4.3. Experimental results and analysis

(1) Hyperparameter Sensitivity Analysis

This experiment aims to analyze the impact of different combinations of CNN block numbers and Transformer block numbers on the accuracy of the model in

Table 4. Model parameter settings.

Module	Parameter	Value
LFE	Channels (Blocks 1, 2, 3)	32,64,128
GFE	MHSA heads (per block)	4
Fusion	Fusion mechanism	Element-wise addition with learnable weights
	Normalization	LayerNorm
Classification	Fully connected layer	4
	Optimizer	Adam
Hyper parameters	Learning rate initialization	0.001 (Cosine annealing dynamic adjustment)
	Batch size	64
	Epochs	100

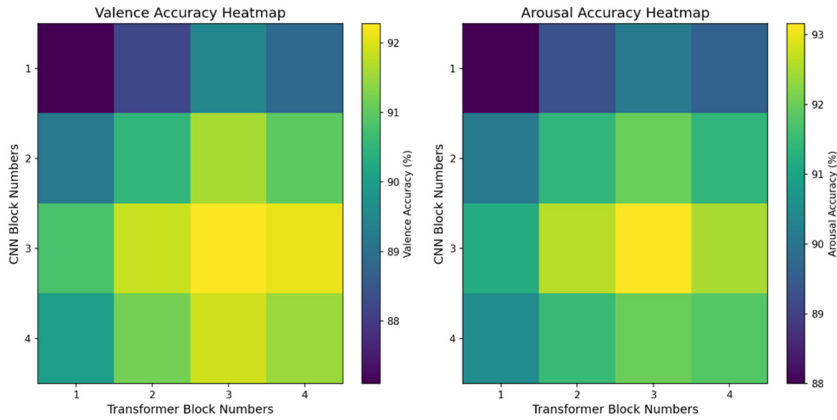


Fig. 3. Impact of CNN and transformer block numbers on valence and arousal accuracy.

valence and arousal classification tasks. By testing various combinations on the DEAP dataset, the optimal block numbers were determined to maximize classification performance. The values for CNN and Transformer block numbers ranged from 1 to 4. The experimental findings are presented in Fig. 3.

The experimental findings demonstrate that the interplay between the amount of CNN blocks and Transformer blocks substantially influences the classification accuracy of valence and arousal. The overall trend shows that as the number of blocks increases, classification performance gradually improves. When CNN blocks = 3 and Transformer blocks = 3, the highest valence classification accuracy of 92.28% is achieved, while arousal classification accuracy reaches a near-peak of 93.16%. However, further increasing the number of blocks leads to a slight decrease in accuracy, potentially due to excessive model complexity, which may introduce feature redundancy or overfitting. Therefore, selecting a moderate number of blocks enhances performance while avoiding unnecessary computational overhead, making it an optimal design choice. The results also reveal that the performance trends for valence and arousal classification are generally consistent, suggesting that the collaborative optimization of local and global features positively contributes to both emotional dimensions.

(2) Model Performance Analysis

To analyze the performance of the proposed model, the following comparison models were selected: CNN,²¹ CNN-LSTM,²² Transformer,²³ temporal convolutional network (TCN),²⁴ and hybrid CNN-transformer.²⁵ Table 5 and Fig. 4 illustrate the experimental findings.

Based on the experimental data in Table 5 and the visual representation in Fig. 4, the following analysis can be drawn:

Valence Dimension. The proposed model (Ours) achieved an accuracy of 92.28% in valence classification, significantly outperforming other models. This indicates

Table 5. Comparison of four-class classification results for valence and arousal dimensions on DEAP across different models.

Model	Valence-Acc (%)	Valence-F1 (%)	Arousal-Acc (%)	Arousal-F1 (%)
CNN	85.12	84.75	87.34	86.90
CNN-LSTM	86.79	85.92	88.41	87.65
Transformer	89.34	88.90	90.12	89.58
TCN	88.67	88.12	89.87	89.23
Hybrid CNN-transformer	90.45	90.12	93.41	93.00
Ours	92.28	91.85	93.16	93.35

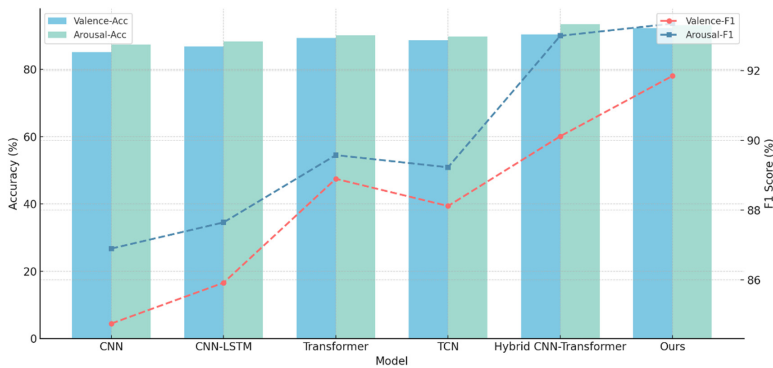


Fig. 4. Comparison of models on DEAP dataset (valence and arousal).

that the proposed model demonstrates superior discriminative ability in recognizing and distinguishing different emotional valences (such as positive or negative emotions). In comparison, the hybrid CNN-transformer model ranked second with an accuracy of 90.45%, showing strong competitiveness. This advantage is likely attributed to the effective integration of the Transformer’s ability to capture long-range dependencies and CNN’s capability in extracting local features. Traditional models like CNN and CNN-LSTM exhibited lower performance in valence classification, with accuracies of 85.12% and 86.79%, respectively, falling short of Transformer-based models. In terms of F1-score, the proposed model also led with 91.85%, reinforcing the fact that the model excels not only in accuracy but also in balancing positive and negative samples. The hybrid CNN-transformer model followed closely with an F1-score of 90.12%, outperforming other models by approximately 1.2% to 5%. The Transformer and TCN models demonstrated comparable F1-scores of 88.90% and 88.12%, highlighting the stability of deep networks in handling emotion classification tasks.

Arousal Dimension. In the arousal dimension, the hybrid CNN-transformer model achieved the highest accuracy at 93.41%, slightly surpassing the proposed model’s 93.16%. This advantage can be attributed to the Transformer mechanism’s superior performance in handling time-series data, effectively capturing the dynamic changes

in emotional arousal. The CNN component enhances the model’s sensitivity to local features, allowing the hybrid model to excel in arousal tasks.

The Transformer and TCN models showed comparable performance in arousal classification, with accuracies of 90.12% and 89.87%, respectively. While these values were higher than CNN (87.34%) and CNN-LSTM (88.41%), they did not surpass the hybrid models. In terms of F1-score, the proposed model achieved 93.35%, slightly higher than the hybrid CNN-transformer’s 93.00%, suggesting that the proposed model can better balance positive and negative samples in arousal tasks, thereby mitigating performance degradation in scenarios with class imbalance. The Transformer and TCN models reported F1-scores of 89.58% and 89.23%, demonstrating high robustness but falling slightly behind the proposed and hybrid models in complex tasks.

(3) Ablation Study Analysis

To verify the impact of each module on emotion classification, an ablation study was conducted. In this experiment, specific modules (such as the feature fusion module, CNN branch, or Transformer branch) were gradually removed or individually activated to assess their contribution to model performance. This experiment helps clarify the role of each module in extracting local and global features, and analyzes their influence on classification accuracy and feature representation. Through this process, the effectiveness of the model’s structural design is validated. The results of the ablation study are presented in Table 6.

The ablation study’s findings suggest that the LFE module is capable of extracting local features with a high degree of accuracy, attaining accuracies of 85.74% for valence and 87.65% for arousal when used independently. However, it lacks the ability to model long-term dependencies. The GFM module performs better in GFM, increasing classification accuracy to 87.21% for valence and 89.02% for arousal, but it shows limited sensitivity to local details. When the feature fusion module was removed, the model concatenated the output features from the CNN and Transformer branches directly to complete the classification task. Although valence accuracy improved to 89.65%, the absence of a dynamic feature interaction mechanism prevented the model from fully exploiting the potential relationships between local and global features, resulting in an F1-score approximately 2% lower than that of the complete model. The full model achieved the highest accuracies of 92.28% for valence and 93.16% for arousal, confirming the critical role of the feature fusion module in integrating local and global features and enhancing classification performance.

Table 6. Ablation experiment results.

LFE	GFM	Fusion	Valence-Acc (%)	Valence-F1 (%)	Arousal-Acc (%)	Arousal-F1 (%)
✓			85.74	85.32	87.65	87.21
	✓		87.21	86.84	89.02	88.67
✓	✓		89.65	89.12	91.37	90.94
✓	✓	✓	92.28	91.85	93.16	93.35

5. Conclusion

This paper proposes an emotion classification model for EEG signals based on multimodal deep feature fusion, aiming to collaboratively model local time-frequency patterns and global contextual dependencies by integrating CNN and Transformers. Extensive experiments on the DEAP dataset demonstrate that the proposed model achieves superior performance compared to several strong baselines, including CNN, CNN-LSTM, Transformer, TCN, and hybrid CNN-transformer models. Specifically, it reaches a valence classification accuracy of 92.28% and an arousal accuracy of 93.16%, along with an F1-score of 93.35%, indicating its effectiveness in capturing both stable and dynamic emotional fluctuations from EEG data. The ablation study further validates the contribution of each model component, showing that the synergy between the LFE and GFM modules plays a critical role in enhancing classification performance. The proposed fusion module enables deep integration of local and global features through dynamic interaction and aggregation, contributing to improved accuracy and generalization. Despite the promising results, the model presents limitations in terms of computational cost. The Transformer branch's MHSA mechanism incurs high resource consumption, which may hinder its application in real-time or resource-constrained environments. Future research will explore model compression techniques such as knowledge distillation and pruning to improve efficiency and deployability. Moreover, the framework can be extended to multimodal emotion recognition by incorporating additional physiological signals, aiming to further improve the robustness and precision of emotion classification systems. Such advancements could provide valuable support for personalized game design and adaptive human-computer interaction.

Acknowledgment

This work was supported by the 2023 General Project of Philosophy and Social Science for Universities in Jiangsu Province (Research on Neural Network Design Platform of Industrial Software for Small Data-driven Enterprise Smart Brain, Project No. 2023SJYB0429) and 2023 Key Project of Jiangsu Graduate Education and Teaching Reform (Research and Construction of “Design +” Graduate Talent Cultivation System Combining Art and Engineering, Project No. 2023YJYJG02).

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References

1. da Silva Franco RY, Santos do Amor Divino Lima R, Monte Paixão RD, Resque dos Santos CG, Serique Meiguins B, Uxmood — a sentiment analysis and information visualization tool to support the evaluation of usability and user experience, *Information* **10**(12):366, 2019.
2. Nunez JC, Cabido R, Pantrigo JJ, Montemayor AS, Velez JF, Convolutional neural networks and long short-term memory for skeleton-based human activity and hand gesture recognition, *Pattern Recognit* **76**:80–94, 2018.
3. Liu Y, Ding Y, Li C, Cheng J, Song R, Wan F, Chen X, Multi-channel EEG-based emotion recognition via a multi-level features guided capsule network, *Comput Biol Med* **123**:103927, 2020.
4. Kim Y, Choi A, EEG-based emotion classification using long short-term memory network with attention mechanism, *Sensors* **20**(23):6727, 2020.
5. Guo JY, Cai Q, An JP, Chen PY, Ma C, Wan JH, Gao ZK, A transformer based neural network for emotion recognition and visualizations of crucial EEG channels, *Phys A: Stat Mech Appl* **603**:127700, 2022.
6. Jafari M, Shoeibi A, Khodatars M, Bagherzadeh S, Shalbaf A, García DL, Acharya UR, Emotion recognition in EEG signals using deep learning methods: A review, *Comput Biol Med* 107450, 2023.
7. Furman L, Duch W, Minati L, Tolpa K, Short-time Fourier transform and embedding method for recurrence quantification analysis of EEG time series, *Eur Phys J Spec Top* **232**(1):135–149, 2023.
8. Almanza-Conejo O, Almanza-Ojeda DL, Contreras-Hernandez JL, Ibarra-Manzano MA, Emotion recognition in EEG signals using the continuous wavelet transform and CNNs, *Neural Comput Appl* **35**(2):1409–1422, 2023.
9. Tharwat A, Independent component analysis: An introduction, *Appl Comput Inform* **17**(2):222–249, 2021.
10. Priyasad D, Fernando T, Denman S, Sridharan S, Fookes C, Affect recognition from scalp-EEG using channel-wise encoder networks coupled with geometric deep learning and multi-channel feature fusion, *Knowl-Based Syst* **250**:109038, 2022.
11. Chowdary MK, Anitha J, Hemanth DJ, Emotion recognition from EEG signals using recurrent neural networks, *Electronics* **11**(15):2387, 2022.
12. Meng M, Zhang Y, Ma Y, Gao Y, Kong W, EEG-based emotion recognition with cascaded convolutional recurrent neural networks, *Pattern Anal Appl* **26**(2):783–795, 2023.
13. Zhong MY, Yang QY, Liu Y, Zhen BY, Xie BB, EEG emotion recognition based on TQWT-features and hybrid convolutional recurrent neural network, *Biomed Signal Process Control* **79**:104211, 2023.
14. Wang Z, Wang Y, Hu C, Yin Z, Song Y, Transformers for EEG-based emotion recognition: A hierarchical spatial information learning model, *IEEE Sens J* **22**(5):4359–4368, 2022.
15. Sharma A, Nigam J, Rathore A, Bhavsar A, EEG classification for visual brain decoding with spatio-temporal and transformer based paradigms, *Proc 5th Indian Conf Computer Vision Graphics and Image Processing*, ACM, pp. 1–9, 2024.
16. Cheng Z, Bu X, Wang Q, Yang T, Tu J, EEG-based emotion recognition using multi-scale dynamic CNN and gated transformer, *Sci Rep* **14**(1):31319, 2024.
17. Liu J, Wu H, Zhang L, Zhao Y, Spatial-temporal transformers for EEG emotion recognition, *Proc 6th Int Conf Advances in Artificial Intelligence*, pp. 116–120, 2022.
18. Chakravarthi B, Ng SC, Ezilarasan MR, Leung MF, EEG-based emotion recognition using hybrid CNN and LSTM classification, *Front Comput Neurosci* **16**:1019776, 2022.

19. Narsimha Reddy CH, Mahesh S, Manjunathachari K, Hybrid feature integration model and adaptive transformer approach for emotion recognition with EEG signals, *Comput Methods Biomech Biomed Eng* **27**(12):1610–1632, 2024.
20. Kulkarni D, Dixit VV, Hybrid classification model for emotion detection using electroencephalogram signal with improved feature set, *Biomed Signal Process Control* **100**:106893, 2025.
21. Li C, Lin X, Liu Y, Song R, Cheng J, Chen X, EEG-based emotion recognition via efficient convolutional neural network and contrastive learning, *IEEE Sens J* **22**(20):19608–19619, 2022.
22. Huang Z, Ma Y, Wang R, Li W, Dai Y, A model for EEG-based emotion recognition: CNN-BI-LSTM with attention mechanism, *Electronics* **12**(14):3188, 2023.
23. Yao X, Li T, Ding P, Wang F, Zhao L, Gong A, Fu Y, Emotion classification based on transformer and CNN for EEG spatial-temporal feature learning, *Brain Sci* **14**(3):268, 2024.
24. Zakiieldin K, Khattab R, Ibrahim E, Arafat E, Ahmed N, Hemayed E, ViTCN: Hybrid vision transformer with temporal convolution for multi-emotion recognition, *Int J Comput Intell Syst* **17**(1):64, 2024.
25. Lu W, Ma H, Tan TP, CIT-EmotionNet: CNN interactive transformer network for EEG emotion recognition, arXiv:2305.05548, 2023.